Legible Consensus: Capability Gaps in Flexible Quorums

Paper #98

Abstract

Distributed-systems monitoring tells operators which nodes are healthy, but not whether a quorum-based service can acquire proposal authority or complete a commit. Phase symmetry makes these capabilities identical; with majority quorums, a reachable-node count therefore answers both questions. Flexible quorum designs can spend that coincidence to make the commit path fast. We characterize the resulting capability gaps exactly. An acquisition-but-not-commit state is impossible if and only if every Phase 1 quorum contains a Phase 2 quorum; the dual gap is impossible under the mirrored containment. Thus, when the induced phase predicates differ, at least one gap exists: directionally ordered predicates admit one direction, while incomparable predicates admit both. In pre-registered single-decree simulations, the two gaps have sharply different consequences. An acquisition-only incumbent was indistinguishable from a healthy contender on every recorded metric and injected accepted values that subsequent acquisition must preserve. Its commit-capable but acquisition-incapable dual prevented a healthy proposer from deciding in all 50 seeds at retry budget eight under the modeled retry policy. We apply the characterization to a wall-shaped quorum construction for physically tiered networks. The wall anchors commits to the fast tier and encodes per-tier participation policy in acquisition. No Phase 2 threshold closes both gaps at every tier, but an $O(\text{tiers})$ readout exposes each tier's capability state and failed obligation from a connectivity summary. A 10-node Earth/LEO/Moon/Mars topology makes the distinction physically undeniable. The same characterization applies to edge and control-plane systems whose effective participation changes faster than logical membership: Mars does not create the gap; it stretches it until node counts, timeouts, and green dashboards can no longer hide it. Quorum properties are verified exhaustively in TLA+; all results are design-level.

1 Introduction

An operator facing a degraded quorum service needs answers to two questions: can some proposer acquire proposal authority, and can it complete a commit? A node-health dashboard answers neither. It reports participants individually, while both capabilities are predicates over the reachable set and the configured quorum families. Treating a green-node count as a service-capability test works only when those predicates happen to coincide.

Planetary distance makes the hidden distinction impossible to ignore. During a Mars conjunction, a set of healthy nodes may remain visible while light travel and a scheduled cut determine which phase obligations can actually be met. Yet Mars is a magnifying glass, not the intended boundary of application. A cloud region, metro aggregation layer, and intermittently connected remote edge form the same latency-ordered structure; backhaul loss, mobility, radio coverage, power cycling, and orchestration churn change effective participation on shorter and less predictable timescales. Mars does not create the capability gap; it stretches the gap until familiar operational approximations—timeouts, node counts, and green health dashboards—can no longer hide it. The terrestrial mapping is a structural applicability argument, not an evaluated result in this paper.

The approximation has a precise historical source. If the same quorum family is used for Phase 1 and Phase 2, then authority acquisition and commit formability are the same predicate over every connectivity state. Majority quorums are the familiar instance, but majority is not the cause; phase symmetry is. Flexible Paxos [9] retains only cross-phase intersection for safety, allowing a rare Phase 1 to be made expensive so that the Phase 2 hot path stays local and fast. That frequency arbitrage spends the operational coincidence that one reachability count once provided.

We characterize the result exactly. Let $R_1(C)$ and $R_2(C)$ say that reachable set C contains a Phase 1 or Phase 2 quorum. We call the mixed states $(R_1, R_2) = (1,0)$ and $(0,1)$ *capability gaps*. The $(1,0)$ gap is empty exactly when every Phase 1 quorum contains a Phase 2 quorum; the $(0,1)$ gap is empty under the mirrored containment. The induced predicates therefore have four mutually exclusive relations: equality admits neither gap; either strict implication admits exactly its opposite-direction gap; and incomparable predicates admit both. Thus differing families do not in general select exactly one gap—their predicate order determines whether one or both are reachable.

The directions are not operationally interchangeable. In our pre-registered single-decree simulations, an acquisition-only incumbent was indistinguishable from a healthy contender on every recorded metric and injected accepted values that a subsequent acquisition had to preserve. In the dual state, a commit-capable but acquisition-incapable incumbent prevented a healthy proposer from deciding in all 50 seeds at retry budget eight under the modeled retry policy. These are bounded observations, not intrinsic labels: the first state is not inherently harmless, and the second result is retry-budget exhaustion rather than a claim of unconditional livelock.

We apply the characterization to a wall-shaped quorum construction over a 5/1/1/3 Earth/LEO/Moon/Mars topology. Phase 2 remains on the fast Earth tier, while each tier's Phase 1 family reads downward through the wall. The upper-tier witnesses are participation policy, not additional Paxos safety requirements: they are useful where local representation, administrative control, sovereignty, or fencing policy requires them. Without such a requirement, the wall is an analytical case study rather than a recommended deployment geometry. With one, its structure exposes which obligation failed and which capability remains, including the difference between scheduled planetary ephemerality and the edge's unscheduled ephemerality.

This paper makes three contributions:

1. **An exact characterization and generic auditor.** We prove the four-way correspondence between quorum-predicate ordering and reachable capability gaps, and provide a construction-independent auditor with deterministic witnesses and an independent exhaustive self-check.

2. **Measured consequences of both gaps.** Pre-registered experiments isolate the sharply different behavior above and report it within the simulator's retry-policy and single-decree bounds (Section 6.4).

3. **A legible, physically tiered case study.** The wall construction exposes per-tier acquisition and commit state in $O(\text{tiers})$ time from a connectivity summary, separates participation policy from anchor safety, and makes the structural edge applicability concrete without presenting it as terrestrial evidence.

Each empirical claim is tied to an executable artifact and output file (Appendix A). All results are design-level; deployment would introduce orbital dynamics, antenna scheduling, and variable link quality not modeled here.

2 Background

2.1 Flexible Paxos

Classic Paxos [11] uses majority quorums for both phases, which are Phase 1 (prepare/promise) and Phase 2 (accept/accepted), guaranteeing intersection via the pigeonhole principle.

Flexible Paxos [9] observed that the intersection requirement is *cross-phase*, not within-phase: any Phase 1 quorum must intersect any Phase 2 quorum, but Phase 1 quorums need not intersect each other, nor Phase 2 quorums each other. Formally, safety holds if $q_1 + q_2 > |N|$, where q_1 and q_2 are the quorum sizes for Phase 1 and Phase 2 respectively. This is the uniform special case where all quorums have the same size; the crumbling-wall construction generalizes this with per-tier quorum families of different sizes, where the cross-intersection property holds by construction rather than by arithmetic majority. The key consequence is that Phase 2 (the hot path, executed on every commit) can be small if Phase 1 (the cold path, executed only when proposal authority is acquired—leader election, under Multi-Paxos) is correspondingly large.

2.2 Crumbling-Wall Quorum Systems

Peleg and Wool [16] introduced crumbling walls as quorum constructions that achieve high availability through asymmetric structure. Nodes are arranged in rows of potentially different widths. A quorum takes one complete row together with one representative from every row below it. The "crumbling" refers to rows losing members to failure.

The key property: any two quorums intersect. If their complete rows differ, the quorum whose complete row sits higher carries a representative in the other quorum's complete row; if they share a complete row, they share it outright—the intersection that consensus requires.

2.3 Composing the Two Ideas

We decompose wall-like structure *across* the two phases rather than reproducing Peleg and Wool's quorums: each tier's Phase 1 family reads downward through the wall from that tier's position, while Phase 2 is the bottom row alone. Only the anchor intersection is safety-critical. The upper-tier witnesses exist to make authority and degradation topology-legible, not to secure agreement—and, as Section 3.6 shows, they are also what manufactures this construction's residual capability exposure.

3 Construction

3.1 System Model

For each configuration epoch e, let N_e be the fixed logical acceptor universe and let $C_t \subseteq N_e$ be the acceptors effectively reachable at time t. A *decision window* is the interval over which we evaluate phase formability against one such connectivity state; N_e remains fixed within that window even when C_t changes between windows. Our results apply epoch by epoch. Changing N_e is a reconfiguration and state-transfer decision that requires its own protocol; it is outside this paper's scope and is not modeled as ordinary reachability churn. For the evaluated epoch below, we write N for N_e.

Let the acceptor universe be

$$N = E \cup L \cup U \cup M,$$

where the tiers are pairwise disjoint and represent Earth, LEO, Moon, and Mars respectively. In the evaluated topology,

$$|E| = 5, \quad |L| = 1, \quad |U| = 1, \quad |M| = 3, \quad |N| = 10.$$

We call this the 5/1/1/3 topology. Tiers are ordered by latency: Earth is the fastest (bottom of the wall), Mars is the slowest (top). The network is asynchronous with configurable per-link delay and jitter. Acceptors are crash-stop and non-Byzantine. Links may be removed during scheduled blackout windows.

The simulator uses three logical consensus scopes:

- **Earth-local**: Flexible Paxos over the five Earth nodes ($q_1 = 4$, $q_2 = 2$).
- **Mars-local**: majority quorum over the three Mars nodes ($q = 2$).
- **Global**: the crumbling-wall construction defined below, over all ten nodes.

3.2 Per-Tier Phase 1 Families

The wall maps tiers to rows, ordered bottom (Earth, fast) to top (Mars, slow), as shown in Figure 1. We index tiers from the bottom: $T_0 = E$ (Earth), $T_1 = L$ (LEO), $T_2 = U$ (Moon), $T_3 = M$ (Mars). A proposer at tier i forms its Phase 1 quorum by reading *downward* from row i: it needs at least one respondent from its own tier and every tier below.

Initiating tier	Phase 1 scope	Tiers	Min size
Earth (tier 0)	Earth only	1	1
LEO (tier 1)	LEO + Earth	2	2
Moon (tier 2)	Moon + LEO + Earth	3	3
Mars (tier 3)	all four tiers	4	4

Let $\mathcal{Q}_1^{(i)}$ denote the Phase 1 family for a proposer at tier i:

$$\mathcal{Q}_1^{(i)} = \left\{ Q \subseteq N \,\middle|\, \forall j \leq i : \; Q \cap T_j \neq \emptyset \right\}.$$

A proposer at tier i needs at least one node from every tier at or below it ($j \leq i$). For example, $\mathcal{Q}_1^{(2)}$ (Moon) requires intersection with T_0 (Earth), T_1 (LEO), and T_2 (Moon), which is every tier from Moon down to Earth. For relaxed Phase 2 (Section 7), Phase 1 additionally requires enough Earth nodes to satisfy a pigeonhole intersection constraint.

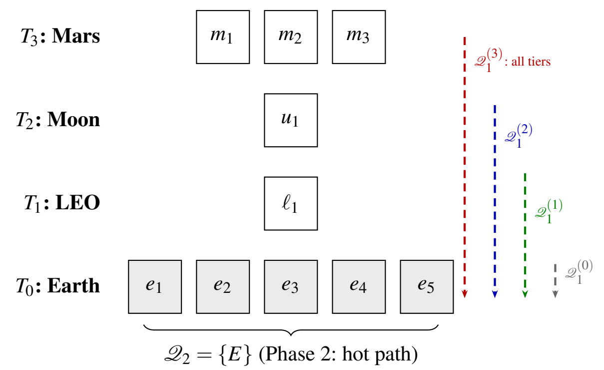

Figure 1: *The crumbling-wall construction over the 5/1/1/3 topology. Each tier's Phase 1 quorum reads downward through the wall (dashed arrows). Phase 2 uses only the Earth row (shaded). The wall narrows from bottom to top: wider rows offer more quorum choices.*

3.3 Phase 2 Family

The Phase 2 family places the hot commit path entirely on the fast tier:

$$\mathcal{Q}_2 = \{E\}.$$

That is, Phase 2 requires all five Earth nodes (the strict construction). Section 7 relaxes this to k-of-$|E|$.

3.4 Safety: Quorum Intersection

Proposition 1. For every tier i and every $Q_1 \in \mathcal{Q}_1^{(i)}$, $Q_2 \in \mathcal{Q}_2$: $Q_1 \cap Q_2 \neq \emptyset$.

Proof. Every $\mathcal{Q}_1^{(i)}$ requires $Q_1 \cap T_0 \neq \emptyset$ (since $0 \leq i$ for all i, i.e., Earth is at or below every tier). $Q_2 = E = T_0$. Therefore $Q_1 \cap Q_2 \supseteq Q_1 \cap T_0 \neq \emptyset$. □

The intersection follows from the wall's structure: every path through the wall reaches the bottom row, and Phase 2 is the bottom row.

We verified this mechanically in TLA+. For readers who have not used it: TLA+ is a specification language whose model checker, TLC, enumerates *every* state reachable from the initial conditions of a finite model and checks the stated invariant in each one. A run is *complete* when that enumeration reaches a fixed point—no reachable state was left unexamined—so a complete run is a proof of the invariant for that finite model, not a statistical sample of behaviors. The `QuorumIntersection` specification enumerates every (construction $\times$ initiating tier $\times$ Q_1 $\times$ Q_2) combination for the strict, k=4, and k=3 families over the full 10-node topology (27,921 states, complete); `ExhaustiveIntersection` re-checks the strict and k=4 families as parse-time assumptions. No intersection failure was found.

For the full Paxos protocol (not just quorum intersection), we verify safety over a reduced 6-node topology ($3E + 1L + 1U + 1M$) that preserves the quorum-intersection structure. The invariant the reduction preserves is the pigeon-

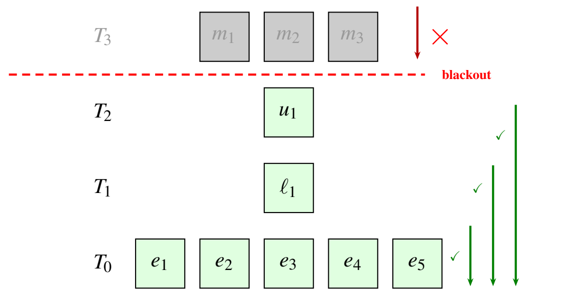

Figure 2: *Liveness under Mars conjunction blackout. The partition (dashed line) disconnects T_3. Earth, LEO, and Moon can still read downward to form Phase 1 quorums (green arrows). Mars cannot reach any inner tier (red $\times$). The wall crumbles from the top.*

hole inequality, not tier membership alone: in the reduced model both quorum families guarantee at least two of the three Earth nodes ($2+2>3$), and in the full modeled family Q_1 carries at least two Earth nodes against a 4-of-5 Q_2 ($2+4>5$)—both instantiate $|Q_1 \cap E| \geq |E|-k+1$ with the margin $|E|-k=1$ held fixed. (Tier membership alone suffices only under strict $\mathscr{Q}_2 = \{E\}$, where any Earth node in Q_1 intersects it.) Liveness properties (crash tolerance, progress under partial failure) are topology-size-dependent and are evaluated only at full scale. Over the reduced topology, single-decree Paxos agreement was model-checked completely (67M states, no violation).

Verification scope. The protocol-level specification models a single all-tier Phase 1 family rather than the per-tier families of Section 3: it exercises the protocol machinery under wall-shaped quorums, but safety of the per-tier construction itself rests on Proposition 1 and the standard Flexible Paxos argument, with the intersection premise checked exhaustively above. The proof is primary; the model checking is corroboration. Specifications are in `tla/` (see Appendix B).

3.5 Liveness: Reading the Wall

The wall determines liveness as follows. A proposer at tier i can complete Phase 1 if and only if it can reach at least one node in every tier j where $j \leq i$ (from its own tier down to Earth). Phase 2 succeeds if the proposer can reach enough Earth nodes (all five under strict $\mathscr{Q}_2$).

During Mars conjunction blackout, cross-tier links between Mars and the inner system are removed: Mars is partitioned from Moon, LEO, and Earth, while those three remain mutually connected. Reading the wall against that connectivity takes one pass per tier. Earth needs only T_0 and is trivially satisfied; LEO needs T_0 and T_1; Moon needs T_0 through T_2; all three are reachable, so all three work. Mars needs a witness from every tier including the inner ones it cannot reach, and is blocked. The liveness failure is scoped to exactly the disconnected tier: the wall crumbles from the top (Figure 2).

This instantiates the legibility definition of Section 1: for any tier i, global Phase 1 succeeds if and only if every tier $j \leq i$ is reachable from i in C—the promised $O(\text{tiers})$ check. This is what distinguishes legibility from mere computability. The claim concerns quorum formability: it bounds what a sole, timely proposer can achieve under connectivity state C, and does not speak to contention between concurrent proposers, whose durable promises can delay an otherwise-capable peer. Global *operation*, as opposed to Phase 1 recovery, additionally requires the Phase 2 quorum to be reachable; Section 6.3 shows the two conditions coming apart.

3.6 Capability States: Where the Two Conditions Come Apart

Phase 1 and Phase 2 formability are two predicates over the same connectivity state, and the wall does not force them to agree. Write $R_1(i,C)$ for "a proposer at tier i can form some $Q_1 \in \mathscr{Q}_1^{(i)}$ under connectivity state C" and $R_2(i,C)$ for "it can form some $Q_2 \in \mathscr{Q}_2$". We say the proposer can *acquire* when R_1 holds and can *commit* when R_2 holds. Under Multi-Paxos, acquisition is leader election and amortizes over many commands; under single-decree Paxos it is per-decree ballot acquisition and every decree pays for it. Our measurements are single-decree throughout (Section 6.4), so "acquisition" carries the general sense everywhere except Section 9, which discusses Multi-Paxos leadership explicitly. The pair (R_1, R_2) admits four joint states. Two are familiar: $(1,1)$ is ordinary operation, and $(0,0)$ is the cut-off tier confined to its local replica. The mixed states are the ones structure alone can hide. In $(1,0)$ a proposer can acquire authority but cannot commit; in $(0,1)$ it can commit but cannot acquire.

Under a single quorum family used for both phases—majority being the familiar case—both mixed states are empty in every connectivity state, because the two predicates are literally the same predicate. This is why counting reachable nodes was ever a sufficient diagnostic. Any construction that makes the phase families differ gives that property up, and which mixed state it admits is determined by the direction in which they differ.

Proposition 2 (containment). $(1,0)$ is unreachable at tier i if and only if every $Q_1 \in \mathscr{Q}_1^{(i)}$ contains some $Q_2 \in \mathscr{Q}_2$. Dually, $(0,1)$ is unreachable at tier i if and only if every $Q_2 \in \mathscr{Q}_2$ contains some $Q_1 \in \mathscr{Q}_1^{(i)}$.

Proof sketch. Both predicates are monotone in C: adding a reachable node never revokes a capability. If some Q_1 contains no Q_2, take C to be exactly that Q_1; then R_1 holds and R_2 fails. Conversely if every Q_1 contains a Q_2, then any C admitting a Q_1 admits that contained Q_2. The dual argument is symmetric. □

Proposition 3 (boundary). For the construction of Section 3 under relaxed $\mathscr{Q}_2 = k\text{-of-}|E|$ (Section 7), $(1,0)$ is unreachable at *every* tier if and only if $|E|-k+1 \geq k$, that is $k \leq \lceil |E|/2 \rceil$. Sufficiency is unconditional—it does not depend on the number of tiers, their sizes, or the initiating tier. The converse additionally requires non-degeneracy: no empty wall row.

The arithmetic is the same hitting-set bound that carries safety. A Q_1 holds at least $|E|-k+1$ Earth nodes and a Q_2 holds k; when $|E|-k+1 \geq k$, every Q_1 already contains a full Q_2 and Proposition 2 applies. For a threshold family used uniformly, with sizes q_1 and q_2, the same reasoning collapses to a statement about two integers: $(1,0)$ is reachable exactly when $q_1 < q_2$, $(0,1)$ exactly when $q_2 < q_1$, and both are empty exactly when $q_1 = q_2$.

The gradient. Proposition 3 governs $(1,0)$ through the Earth floor. It says nothing about $(0,1)$, which on this construction arises from a different mechanism: the downward chain. A proposer at T_2 (Moon) needs one witness from T_1 (LEO), and this topology has exactly one LEO node. Sever it while all five Earth nodes remain reachable and $\mathscr{Q}_2$ is formable while $\mathscr{Q}_1^{(2)}$ is not. Exhaustive enumeration over all 2^{10} connectivity states, for every tier and every k, gives:

	T_0 (E)	T_1 (L)	T_2 (U)	T_3 (M)
$(1,0)$ reachable	$k \geq 4$	$k \geq 4$	$k \geq 4$	$k \geq 4$
$(0,1)$ reachable	$k \leq 2$	$k \leq 2$	all k	all k

Read the two rows together: *no value of k closes both states*. At $k \leq 2$ the Earth floor exceeds k and every tier carries $(0,1)$ exposure. At $k \geq 4$ Proposition 3's bound is violated and $(1,0)$ opens everywhere. At $k = 3$—the best available configuration, and the relaxation Section 7 adopts—$(1,0)$ is closed at every tier and $(0,1)$ is closed at Earth and LEO, but Moon and Mars retain it. Those are exactly the tiers with an intervening row between themselves and the anchor, and no choice of k removes it, because the obligation that fails is a witness row rather than an Earth count.

The enumeration credits each proposer with a reachable node in its own tier, modelling a colocated acceptor; without that assumption LEO also retains $(0,1)$ at every k, since its single node can be severed while Earth remains whole. We state the assumption rather than absorb it: it determines whether phase symmetry is affordable at one tier or two.

Relationship to Flexible Paxos. The $(1,0)$ state is not new, and the Flexible Paxos paper describes it twice [9]: once in the $|Q_1|$=1, $|Q_2|$=N construction, and once in the row-failure discussion, where completing Q_1 "recover[s] all past decisions" before falling back to reconfiguration. Both readings are constructive, and our measurements agree with them (Section 6.4). Our relationship to that work is therefore threefold. It identifies the state and a use for it; we supply the design-time existence criterion—for a given construction, whether the state arises at all, which §4.3 does not ask; and we find that the cost, where it exists, lies in the unexamined dual. What §4.3 assumes is available and the connectivity setting does not supply is a *recognizer*: its prescribed response presumes the system knows which state it occupies. Under a known crash-failure set that is reasonable. Under connectivity it is exactly the missing predicate, and supplying it is what the readout of Section 3.5 does.

This is the sense in which legibility is load-bearing rather than decorative. The gradient is not an artifact we can relax away; it is a property of any wall whose non-anchor rows carry witness obligations. When a capability gap cannot be closed by construction, making it legible is not a consolation—it is the only mitigation left. The connectivity data needed is already available to an operator; what was missing was the predicate that reads it. Enumeration artifacts are in `results/capability/` (see Appendix B).

3.7 What "Global" Means

When an Earth-based proposer completes both phases using only Earth nodes, the decision is *global* in the Paxos sense: any future proposer at any tier, upon completing Phase 1, will discover this value in the Earth acceptors' state and cannot contradict it. Global consensus does not require that all tiers participated, only that no future quorum can produce a conflicting result.

4 Related Work

Paxos established the standard safety argument for replicated state machines under asynchronous messaging and crash failures [11]. Flexible Paxos showed that safety requires only Phase 1/Phase 2 cross-intersection, not universal majority [9]. Our construction builds directly on that result; the novelty is not in the intersection theorem but in mapping the quorum shape to physical topology and characterizing the resulting liveness properties per tier.

Grid and crumbling-wall quorum constructions [3, 16] demonstrate that quorum geometry can be engineered around combinatorial structure. Howard et al. note explicitly that Flexible Paxos can use such constructions; we take up that suggestion. Peleg and Wool analyzed crumbling-wall availability under random, independent node failures. Our contribution is not the wall itself but the mapping: rows correspond to physical latency tiers, and the resulting per-tier liveness properties under scheduled, topology-correlated disconnection, are the properties the abstract framework identifies as possible but does not explore. Li, Chan, and Lesani [12] formalize heterogeneous quorum systems in which each process may use a different quorum family, and give necessary and sufficient conditions for liveness; our per-tier Phase 1 families are a concrete, topology-aligned instantiation of that general idea. Their Satrapy system [13] carries the framework from abstract quorum systems to practical consensus. Hierarchical quorum structures have a longer history in database replication [5] and coordination services; our wall differs in that the hierarchy reflects physical latency tiers rather than organizational or logical grouping.

Geo-distributed consensus provides the closest practical analog. WPaxos [1] uses Flexible Paxos with grid quorum layouts and per-object multi-leader coordination to keep commits local; it is the nearest point of comparison. WPaxos's grid quorums are symmetric: Q_1 spans $Z - f_z$ zones and Q_2 spans $f_z + 1$ zones, with the same structure regardless of which node initiates. Our per-tier $\mathscr{Q}_1^{(i)}$ families are asymmetric as the initiating tier determines the quorum scope, which produces a leadership cost gradient that symmetric grids structurally cannot express (Section 9). EPaxos [15] removes the leader bottleneck for non-conflicting commands. Mencius [14] distributes leader responsibility across WAN sites. MDCC [10] achieves single-round cross-datacenter commits. All four optimize for tens-to-hundreds of milliseconds WAN latency and assume all replicas are usually reachable. Our setting differs in two respects: the asymmetry spans three orders of magnitude (subsecond to 1342 s), and the target phenomenon is scheduled total disconnection rather than heterogeneous latency.

Delay-tolerant networking [2, 7] motivates the operating regime but addresses routing and custody transfer, not consensus. CRDTs have been studied in DTN-like settings [8]. SwarmDAG [18] uses extended virtual synchrony for partition-tolerant robot swarms; Tran et al. [17] measure consensus latency under partitioning. Protocol Labs' IPC [6] uses hierarchical subnets for blockchain scaling. NASA's DSN study [4] evaluates database replication for ground stations. Our work occupies a gap: Paxos safety under topology-shaped quorum geometry and scheduled disconnection, with per-tier liveness characterization.

5 Experimental Design

Eidolon is a discrete-event simulator built on SimPy. Acceptors process Prepare and Accept messages with configurable processing delay. Proposers fan out messages to all acceptors and collect responses with a per-phase timeout. The network model assigns entities to named locations with per-link latency, jitter, and optional loss. Blackout is modeled as deterministic link removal; the repeater model substitutes a degraded alternate path. Event ordering is by simulation timestamp; the simulator is deterministic given a seed.

5.1 Topology

Five Earth ground stations (NA-West, Europe, Asia, SA-East, Africa) with cross-continental latencies of 30–120 ms. One LEO satellite with 20–45 ms links to ground stations. One Moon node at 1.28 s one-way. Three Mars nodes at configurable one-way delay (186–1342 s, representing closest approach to opposition). Mars-local links are 5 ms.

Table 1: *Experimental design.*

Experiment	Purpose	Key parameters
Parameter sweep	Confirm structural invariance of Earth-initiated global consensus across Mars delay and blackout duration	Mars delay $\in \{186, 750, 1342\}$ s; blackout $\in \{300, 900, 1800\}$ s; 50 seeds per point
Per-tier sweep	Measure which tiers retain global liveness; compare sparse vs. full-coverage topologies	Initiating tier $\in$ {Mars, Moon, LEO, Earth}; both topologies; 186 s Mars delay; 900 s blackout; 50 seeds
Crash tolerance	Characterize the tradeoff between Phase 2 relaxation (k-of-$\lvert E\rvert$) and crash tolerance	$k \in \{3, 4, 5\}$; 0–2 Earth crashes; 186 s; 900 s blackout; 50 seeds

We evaluate two network variants:

- **Sparse**: LEO has links to 3 of 5 Earth stations (NA-West, Europe, Asia). Mars has links to 2 of 5 (NA-West, Europe). This models a single LEO satellite with limited ground coverage and a Mars relay with two tracking stations.

- **Full coverage**: LEO and Mars have links to all 5 Earth stations. This models a LEO constellation with global coverage and a deep-space network with worldwide antenna placement.

5.2 Experiments

Four experiment families probe different aspects of the construction; the fourth, the pre-registered valence experiment, is described with its results in Section 6.4.

Fixed parameters across all experiments are listed in Table 6 (Appendix A). All seeds are 40–89 (50 runs per point). Results are reported as means with 95% confidence intervals.

6 Results

6.1 Flat versus Crumbling Wall: Geometry Alone Changes the Outcome

To calibrate the effect of quorum geometry, we compare three constructions over the identical topology, blackout model, and parameter sweep. The *flat* construction uses a single Phase 1 family requiring all four tiers:

$$\mathscr{Q}_1^{\text{flat}} = \{Q \subseteq N \mid \forall j \in \{0,1,2,3\} : Q \cap T_j \neq \emptyset\}.$$

Table 2: *Effect of quorum geometry: flat vs. majority vs. crumbling-wall Phase 1 over the same topology and blackout model. Earth-initiated global proposer, hard blackout, 50 seeds per point; rates are identical at all 9 parameter points (3 Mars delays × 3 blackout durations). Latency is the full two-phase (Phase 1 + Phase 2) decision latency of a single attempt.*

Construction	During	Post	Latency
Flat Phase 1 (all tiers required)	0.0%	varies	varies
Majority Phase 1 (any 6 of 10)	100.0%	100.0%	0.361 s
Crumbling wall (Earth reads Earth only)	100.0%	100.0%	0.183 s

Any blackout that removes a tier makes this quorum unformable. The *majority* baseline is a competitive shape-free Flexible Paxos configuration: $\mathscr{Q}_1^{\text{maj}} = \{Q \mid |Q| \geq 6\}$, any majority of the ten nodes, with no tier structure. The *crumbling wall* construction uses the per-tier Phase 1 families defined in Section 3, with an Earth-based global proposer. All three satisfy cross-intersection with $\mathscr{Q}_2 = \{E\}$: the tiered families require $Q_1 \cap T_0 \neq \emptyset$ directly, and any 6 of 10 nodes must include an Earth node because only five non-Earth nodes exist.

The flat construction yields 0% during-blackout success at every sweep point because it requires all tiers, including disconnected Mars. The crumbling wall yields 100% because the Earth-based proposer's Phase 1 reads only Earth. Zero variance across 50 seeds confirms that outcomes here are structural consequences of quorum geometry and link state, not scheduler artifacts. We therefore treat this as calibration, not discovery.

The majority baseline separates what the wall's geometry buys from what any sufficiently slack quorum buys. Majority Phase 1 also sustains 100% during-blackout success for the Earth-initiated proposer: the blackout removes three nodes and a 6-of-10 quorum tolerates four, so blackout survival here is within the slack of a shape-free quorum. What the geometry changes is everything else. Two-phase decision latency doubles (0.361 s vs. 0.183 s, uniformly across the sweep): the sixth-fastest respondent lies outside Earth, so majority Phase 1 leaves the fast tier even when nothing has failed. Both constructions share the identical all-Earth Phase 2, so steady-state Multi-Paxos commits do not differ; the penalty falls on authority acquisition, recovery, and single-decree decisions—every operation that pays Phase 1. The tolerance is not tier-aware: during blackout a LEO-initiated proposer reaches only five acceptors (itself, three ground stations, and Moon), so majority Phase 1 is unformable from LEO while the wall's LEO family still completes Phase 1. And the valid-quorum count collapses to a single tier-independent value ($\sum_{k\geq 6}\binom{10}{k} = 386$), below Earth's 992 wall quorums and without the per-tier gradient of Figure 3. The wall's contribution over a well-chosen flexible quorum is therefore not raw blackout survival, which slack alone can purchase; it is what slack cannot express—Phase 1 kept inside the fast tier, per-tier legibility of degradation, and the leadership gradient of Section 9.

Table 3: *Per-tier global consensus during Mars conjunction blackout (full-coverage topology, hard blackout, 186 s Mars delay, 900 s blackout, mean ± 95% CI over 50 seeds). Rates are identical across seeds (structural outcomes; no CIs shown); recovery lags are cadence artifacts (see text).* †*Undefined: no Mars attempt lies wholly within this window (see text).*

Tier	Position	During	Latency	Rec. lag
Earth	bottom (0)	100.0%	0.183 s	68.1 s
LEO	tier 1	100.0%	0.131 s	67.2 s
Moon	tier 2	100.0%	5.131 s	27.1 s
Mars	top (3)	—†	752.5 s	1009.3 s

6.2 Per-Tier Liveness: The Wall Crumbles from the Top

Table 3 reports the central result. Under full network coverage, each tier runs its own global proposer during Mars conjunction blackout (186 s one-way, 900 s blackout, hard blackout model, 50 seeds).

Three of four tiers retain global consensus capability during hard Mars blackout. Only Mars, which is the disconnected tier, loses it. The liveness failure crumbles from the top of the wall, exactly as the construction predicts. (95% CIs: latency within ± 0.001 s and recovery lag within ± 0.01 s for Earth, LEO, and Moon; ± 0.7 s and ± 1.7 s for Mars.)

The latency column maps directly to physics. Moon's 5.131 s is two Paxos phases at the 1.28 s one-way Moon–Earth light delay ($\approx 4 \times 1.28$ s). Earth's 183 ms reflects cross-continental round-trips within the Earth tier. LEO's 131 ms is *faster* than Earth, which is a counterintuitive result explained by the network topology: LEO-to-ground links (20–45 ms) are shorter than cross-continental Earth links (50–120 ms). Wall position determines quorum *obligations*; the speed of light determines their *cost*. These are correlated but not identical, and LEO is the proof.

Mars fails during blackout for the structural reason predicted by the wall (it cannot reach inner tiers for Phase 1). At the table's 900 s point, however, no Mars attempt lies wholly within the blackout—a failed attempt occupies one 500 s Phase 1 timeout, and the attempt cadence, phase-shifted by the long pre-blackout round, straddles the boundary—so the during-blackout cell is undefined rather than zero. The prediction is observed directly at the 1800 s blackout duration, where the cadence places complete failed attempts inside the window: during-blackout success is 0% across all 50 seeds. Outside blackout, however, Mars-initiated global consensus *succeeds*: every pre- and post-blackout attempt completes across all 50 seeds, at a mean latency of 752.5 s—two Paxos phases at the 186 s one-way Mars–Earth delay ($\approx 4 \times 186$ s).

Table 4: *Per-tier global consensus under sparse topology (same parameters as Table 3). All rates are identical across the 50 seeds—structural outcomes, reported without CIs.*

Initiating tier	Wall prediction	Sparse result
Earth	works	100.0%
LEO	works	0.0%
Moon	works	100.0%
Mars	blocked	0.0%

Success requires a per-phase timeout budget sized to the initiating tier's physics: the worst-case request–response path from Mars is $2 \times 186 \approx 372$ s, within the 500 s per-phase budget of Table 6. This is not a blackout finding but a physics finding: Mars-initiated global consensus is possible but pays the full interplanetary round trip on every phase, so in practice delegation to a lower-tier proposer dominates.

Recovery lag differences between tiers (27.1 s for Moon vs. 68.1 s for Earth) are cadence-alignment artifacts, not protocol properties. Recovery lag measures the gap from blackout end to the next successful cadence-aligned attempt, so it is quantized by the 120 s reconciliation interval: its value is bounded by one interval plus a round time, and where it falls within that bound depends only on where the blackout boundary lands in each tier's attempt cadence.[1]

6.3 Wall Obligations versus Network Reachability

Table 4 repeats the per-tier experiment under the sparse network topology.

LEO drops from 100% (full coverage) to 0% (sparse). The wall says LEO needs LEO + Earth; that obligation is satisfiable. But strict Phase 2 requires all five Earth nodes, and LEO has links to only three ground stations. The wall determines what a tier *needs*; the network determines what it *can reach*. Liveness requires both.

This makes legibility a two-step operator procedure: check the wall structure (which determines quorum obligations per tier) and check the network topology (which determines whether those obligations are achievable). The wall is the blueprint; the network is the building site.

Moon succeeds in both topologies because it has links to all five Earth ground stations. The sparse topology models a single LEO satellite with limited ground coverage; the three missing ground-station links are the difference between 0% and 100% for LEO-initiated consensus.

[1]Each tier's per-round time shifts its cadence phase relative to blackout end, which is why the values differ across tiers and move when the experiment window shifts. The effect is deterministic and independent of the consensus protocol; the bound, not the value, is the protocol property.

6.4 The Valence of the Two Capability States

Section 3.6 says when each state arises. It does not say what either costs, and the two turn out to cost very different things. We registered predictions, a falsification criterion, and the consequence of a null before the experiment existed; the primary prediction was falsified in the opposite direction from the one expected, and we report it as registered.

The setting is single-decree Paxos. A healthy proposer competes with an incumbent whose capability state is set by a mid-round connectivity change, with the retry budget swept over $\{1,2,4,8\}$ and 50 seeds per cell. The incumbent sits at the Moon tier, since an Earth-tier proposer is colocated with an Earth acceptor and no partition can fail its Phase 1 while leaving Earth reachable. The experiment exists only at k=5: by Proposition 3, at $k \leq 3$ there is no $(1,0)$ state to place an incumbent into.

Incumbent state	$P(\text{ok})$	med. ttfc	rounds	NACKs
$(1,0)$, flip-induced	1.000	3.420 s	2	7
$(1,1)$, fully healthy	1.000	3.420 s	2	7
$(0,1)$	0.000	—	—	48–49

The first two rows are the result. A $(1,0)$ incumbent is metric-for-metric indistinguishable from a perfectly healthy competing proposer, at every retry budget, on every column we recorded. Whatever a $(1,0)$ incumbent costs the system, an ordinary contending peer costs exactly the same: the cost is contention, not the capability state. Its arm is also flat in the retry budget while its own Phase 1 completions climb from 1 to 8. The third row is the state nobody examined: at retry budget 8 the healthy proposer fails in 50 of 50 seeds, with Wilson intervals disjoint from the row above. Where the $(0,1)$ state is placed within the round does not matter—early and late placements overlap at every budget.

Mechanism: failure is cheap, success is expensive. A proposer that fails Phase 1 learns so immediately and retries after a short backoff. A proposer that *completes* Phase 1 and then cannot form $\mathcal{Q}_2$ must wait a full phase timeout before it can try again. It is throttled by its own success. The policy-independent core is that completing a phase imposes a floor on the retry interval—at least one phase timeout, which must exceed the round trip—while failing a phase imposes no floor at all. How small the floor-free cycle is depends on the backoff policy, which we deliberately do not study; the existence of the asymmetry does not.

Two things this is not. The observed blocking is exhaustion of a bounded retry budget, not livelock: we make no livelock claim. And the $(1,0)$ result is not merely an absence of harm. Its partial Phase 2 reaches four of the five Earth acceptors; the next proposer's Phase 1 surfaces that accepted value, and ordinary Paxos value adoption carries it to commitment. The $(1,0)$ proposer is a value injector rather than a spoiler, which is the constructive reading of Flexible Paxos §4.3 observed with its mechanism visible.

Scope. Under Multi-Paxos a $(0,1)$ tier keeps committing while its authority survives, deferring exposure to the moment authority must be re-acquired; under single-decree every decree needs Phase 1, so the block is immediate. The measured setting is the severe one.

Deviations, recorded. Five, with reasons, in the artifact record alongside the verdict table and threats: the flip lands before Phase 2 fan-out rather than in its return leg, which would test acknowledgment loss instead of a capability state; the intended $(0,0)$ control was realized as $(0,1)$, the only state failing Phase 1 while still reaching the anchor; and one arm was added because the registered timing prediction was untestable as written.

7 Crash Tolerance and Coordinated Relaxation

The strict Phase 2 choice $\mathscr{Q}_2 = \{E\}$ introduces a fragility: a single crashed Earth acceptor blocks all global Phase 2 progress. This section relaxes that choice.

7.1 Relaxed Construction

Let the relaxed Phase 2 family require k of $|E|$ Earth nodes. We evaluate $k = 4$ (tolerates one crash) and $k = 3$ (tolerates two). Phase 1 requires correspondingly more Earth nodes to maintain intersection: a k-of-$|E|$ Phase 2 quorum omits at most $|E| - k$ Earth nodes, so any set of $|E| - k + 1$ Earth nodes intersects every Phase 2 quorum (pigeonhole). Phase 1 must therefore contain at least $|E| - k + 1$ Earth nodes. For $k = 4$ out of 5 Earth nodes, this means at least 2 Earth nodes in every Q_1; for $k = 3$, at least 3. Both relaxed constructions are verified exhaustively in TLA+ alongside the strict one (Section 3).

The relaxed Phase 1 family for tier i is:

$$\mathscr{Q}_{1,k}^{(i)} = \left\{ Q \subseteq N \;\middle|\; \forall j \leq i : \, Q \cap T_j \neq \emptyset, \, |Q \cap E| \geq |E| - k + 1 \right\}.$$

7.2 Results: The Weakest Link Migrates

Table 5 reports the crash-tolerance sweep (Earth-initiated, repeater-assisted, 186 s Mars delay, 900 s blackout, 50 seeds).

Strict Phase 2 is completely crash-intolerant. A single Earth crash produces permanent global liveness loss (row 2).

The weakest link migrates. At two crashes with standard Earth-local quorums, the *global* construction ($k = 3$) maintains 100% during-blackout success but the *Earth-local* construction loses liveness outright: it cannot form a Phase 1 quorum of 4 with only 3 remaining Earth nodes, so every attempt after the crashes land fails, and the 60.5% aggregate rate reflects only attempts completed before the crashes (rows 5–6). The global construction's advantage is not that it spans more nodes—for the Earth-initiated proposer used here, both phases of the relaxed global construction touch only Earth nodes—but that its Earth requirements are smaller: at $k = 3$, both global phases need only the three surviving Earth nodes, while the Earth-local configuration still demands four. The structural insight is the legibility of the migration: which crashes matter depends on which quorum family's requirements they intersect, and the wall makes each family's requirements explicit.

Coordinated relaxation restores resilience. Relaxing the Earth-local quorum to majority ($q_1 = q_2 = 3$) restores 100% local success with no loss of global success (row 6). The tradeoff curve, with Phase 1 minima shown for the bottom-initiated (Earth) and top-initiated (Mars) proposers:

Regime	P1 min (E/M)	P2	Tol.	Earth-local P2 quorum
Strict ($k = 5$)	1 / 4	5-of-5	0	2-node E
Relaxed ($k = 4$)	2 / 5	4-of-5	1	2-node E
Relaxed ($k = 3$) + maj	3 / 6	3-of-5	2	3-node E

An Earth-initiated Q_1 needs only the $|E| - k + 1$ Earth nodes; a Mars-initiated Q_1 adds one witness from each of the three upper tiers. Relaxation buys crash tolerance by growing the cold path, never the hot one: Phase 2 shrinks as Phase 1's Earth requirement grows.

8 Threats to Validity and Limitations

All results are design-level, not deployment-level.

Anchor concentration. The wall concentrates both safety and liveness at the bottom row. Every Phase 1 reads down to Earth; every Phase 2 is on Earth. In our construction, this concentration is sufficient for the legibility property: per-tier liveness determination is $O(\text{tiers})$ because a fixed reference point for cross-intersection eliminates the need to enumerate quorum subsets. Distributing the anchor across multiple rows would require reasoning about which row combinations provide intersection—structured multi-anchor families might remain efficiently analyzable, but the wall's simple linear readout would be lost. The single anchor is what makes the wall readable at a glance. Note, however, that the anchor is a *tier*, not a node. Since safety depends only on tier membership (Section 3), the anchor tier's internal replication is unconstrained by the wall—it can be scaled, geo-distributed, or backed by its own consensus protocol independently. The threat is tier-level partition (all of Earth unreachable simultaneously), not individual node failure; relaxed Phase 2 tolerates the latter.

Table 5: *Coordinated relaxation under Earth crashes (mean ± 95% CI, 50 seeds). Earth-local Flexible Paxos: "std" is q_1=4,q_2=2; "maj" is q_1=q_2=3. Recovery is the lag from blackout end to the next successful global attempt and is quantized by the 120 s reconciliation cadence (Section 6).*

Global $\mathscr{Q}_2$	Local	Crashes	Earth-local	Global during	Global post	Recovery (s)
strict	std	0	100.0%	100.0%	100.0%	93.11±0.01
strict	std	1	100.0%	0.0%	0.0%	n/a
4-of-5	std	0	100.0%	100.0%	100.0%	94.27±0.01
4-of-5	std	1	100.0%	100.0%	100.0%	94.26±0.01
3-of-5	std	2	60.5%	100.0%	100.0%	94.64±0.01
3-of-5	maj	2	100.0%	100.0%	100.0%	94.63±0.01

"Tier" is a bundle. A tier in this paper bundles several properties that happen to coincide in the planetary setting: a latency regime, a failure and partition domain, and, implicitly, an administrative domain. They need not coincide in general—two lunar bases operated by different agencies would share a latency regime but not an administrative one, and two datacenters in one administrative domain may sit in different latency regimes. The wall as defined indexes only latency-ordered reachability; decomposing the remaining dimensions is future work.

Abstract network model. Blackout is deterministic link removal. Orbital dynamics, antenna scheduling, contention, signal coding, and variable link quality are not modeled. The LEO topology variant demonstrates that network reachability matters; a more realistic model with time-varying LEO coverage (orbital ground tracks) would show liveness fluctuating with satellite position.

Stylized workload. Reconciliation runs on a fixed 120 s interval. This creates scheduling artifacts in recovery lag measurements; recovery lag differences between tiers are workload properties, not protocol properties.

Single topology. We evaluate one topology size (5/1/1/3). The construction generalizes to other tier sizes, but we do not sweep Earth-tier size or Mars replication degree.

Crash-stop only. Byzantine behavior, correlated faults, reconfiguration during blackout, and storage failures are outside scope.

Per-tier crash tolerance. The interaction between wall position and crash tolerance (does a Moon-initiated proposer degrade differently from an Earth-initiated one under Earth crashes?) is not explored and remains future work.

9 Discussion

Separation of inter-tier obligation from intra-tier replication. The separation introduced in Section 1 has a concrete cost consequence: the wall's per-tier Phase 1 obligation (one witness from the tier) is strictly cheaper than requiring intra-tier consensus. Mars's three nodes are independent witnesses—any one satisfies the wall's requirement. A Mars-internal majority would require two of three, adding a quorum obligation the wall does not need. The anchor tier is the sole exception: Phase 2 coherence requires it to behave as a single logical entity, achievable through internal replication (a locally replicated state machine). Non-anchor tiers benefit from redundancy without paying the cost of within-tier agreement. The TLA+ verification provides independent evidence for this separation: the reduced 6-node topology ($3E+1L+1U+1M$) is not merely a computational shortcut but a model of the collapsed-tier design—shrinking Earth from five nodes to three preserves the quorum intersection properties because the Phase 2 threshold rescales with it, holding the pigeonhole margin $|E|-k=1$ fixed (under strict $\mathscr{Q}_2$, tier membership alone suffices). The same structural invariant that makes the reduction valid makes the deployment collapse cheap.

Topology-scoped consistency. The capability state determines which consistency contract a tier can honestly offer, and the sparse topology of Section 6.3 separates the two capabilities sharply. Sparse LEO completes Phase 1, which surfaces accepted values, but *knowing* a value is chosen requires acceptance evidence from a complete Phase 2 quorum—all five Earth nodes under the strict family—and sparse LEO reaches three. Phase 1 alone buys visibility, not knowledge. The client-facing ladder follows directly: a tier holding both capabilities serves linearizable operations on the global log; a tier that can assemble certification evidence but not acquire proposal authority can serve committed if stale reads; an observe-only tier can return values only under an explicitly uncertified contract, since what it sees may yet be superseded; a cut-off tier is confined to its local scope. Which contract to expose over a degraded capability set is a service-level decision the wall does not make. What it makes legible is the boundary, and therefore when that decision is being taken. The degradation is epistemic before it is procedural: what a tier loses first is not the right to propose but the ability to *know* what has been committed. At interplanetary distances the gap between the global and local regimes is twenty-two minutes wide—a distinction students struggle with as an abstract definition becomes a physical fact you could set a timer by.

Opposite remediations. For Multi-Paxos deployments the two mixed states prescribe contradictory responses to the same symptom, and this is where legibility earns its keep operationally. In $(0,1)$ a live leader is committing normally and no replacement can be elected; the standard reflex of restarting the leader therefore converts a degraded-but-serving system into a stalled one. In most other states that same reflex is harmless or curative. The symptom an operator sees—writes slowing, nothing obviously wrong, every node green—is identical across both, and no node-health summary distinguishes them. The capability readout does.

Leadership hierarchy. The wall imposes a cost gradient on Multi-Paxos leader election that symmetric quorum constructions cannot express. Figure 3 shows the number of valid Phase 1 quorums per tier. The counts admit a closed form: the tier-i family contains $N_i = 2^{f_i}\prod_{j\le i}(2^{|T_j|}-1)$ quorums, where each required tier contributes a nonempty subset and the f_i nodes above tier i are unconstrained—$992 = 31\cdot 2^5$ for Earth down to $217 = 7\cdot 31$ for Mars. The exhaustive TLA+ enumeration confirms these counts rather than defining them. The closed form also makes the monotonicity a lemma rather than an observation: raising the initiator by one tier changes exactly one factor, tier T_{i+1}'s contribution, from unconstrained $(2^{|T_{i+1}|})$ to nonempty $(2^{|T_{i+1}|}-1)$, so $N_{i+1} = N_i\cdot\big(2^{|T_{i+1}|}-1\big)/2^{|T_{i+1}|} < N_i$ for every wall, independent of tier sizes; the sequence $992\to 496\to 248\to 217$ instantiates the ratios $\frac{1}{2},\frac{1}{2},\frac{7}{8}$. The argument survives relaxed Phase 2 unchanged, since the Earth-minimum factor is identical for every initiator tier and cancels from the ratio. The families are moreover nested—every tier-$(i{+}1)$ quorum is also a tier-i quorum—so any crash pattern under which a higher tier can still elect is one under which every lower tier can too; the counts quantify the gap in valid quorum sets, not availability under a stochastic failure model. In Multi-Paxos, where leader election *is* Phase 1, this makes Earth leadership both faster (sub-second vs. minutes) and strictly more crash-pattern-tolerant than Mars. The gradient is expressible only because the families are per-tier: any construction with a single initiator-independent Phase 1 family assigns every tier the same count by definition, which is the precise sense in which symmetric constructions cannot express it.

WPaxos [1] uses symmetric grid quorums with uniform Q_1 structure regardless of initiator. By contrast, the wall uses tier-indexed families $\mathscr{Q}_1^{(i)}$, so partition failover cost is topology-proportional: if the leader is on Earth and Mars disconnects, leader election remains Earth-scoped.

Terrestrial mapping, as a prediction. The same structure appears wherever topology creates latency asymmetry: a cloud region as anchor, metro edge above it, a remote site above that; or disconnected, intermittent, low-bandwidth networks reachable only through windowed links. What makes these more than analogies is the direction of the pressure. Every latency economy argues for shrinking the commit quorum so the hot path stays local, and by the criterion of Section 3.6 that is precisely the direction which opens the

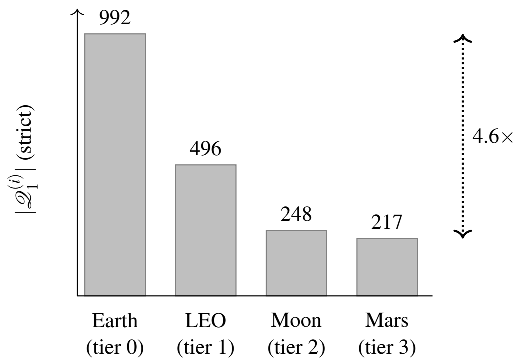

Figure 3: *Phase 1 quorum count by initiating tier (strict $\mathscr{Q}_2$, from exhaustive TLA+ enumeration). Leadership flexibility increases monotonically from top to bottom of the wall.*

commit-capable-but-acquisition-incapable state. The incentive that makes flexible quorums attractive at the edge is the incentive that admits the harmful gap, and it strengthens as deployments spread. We state the status of this plainly: it is an argument from structure, not an evaluated result. Every claim in this paragraph is a hypothesis our criterion makes precise, and testing it on terrestrial topologies is future work.

Future work. The separation principle suggests three natural extensions: scoped write leases, where the anchor tier delegates write authority to upper tiers over bounded slot ranges—lease expiry on disconnection, conflict resolution via MRDTs/CRDTs—so that topology-scoped consistency becomes a managed contract rather than a binary cutoff; structured multi-anchor families that trade part of the linear readout for tolerance of anchor-tier partition (Section 9); and empirical validation of the terrestrial mapping on edge/cloud topologies where the same decomposition applies but the latency asymmetry is less dramatic.

10 Conclusion

A single quorum family used for both phases supplies two guarantees, and the literature named one. Intersection is a witness to the past; coincidence—the two phase predicates agreeing in every connectivity state—is what made counting reachable nodes an adequate diagnostic, and flexible quorums spend it. We give the exact criterion for which capability gap that opens, and our own wall returns no comfortable answer: no threshold setting closes both, and the harmful one persists at every tier separated from the anchor by an intervening row—its witness obligations buying topology-legible authority and manufacturing that exposure in one stroke.

Which is why we take legibility as the result rather than the consolation. The state is invisible to the instrument operators use—every node healthy, nothing concealed, and for Multi-Paxos deployments the standard remediation is the one that makes it worse.

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A Simulator Parameters

B Claim-to-Artifact Traceability

Table 7 maps the paper's main claims to executable artifacts. Paths are relative to the repository root of the (anonymized) artifact accompanying this submission. Results were generated with Python $\geq$3.14 and SimPy $\geq$4.1.1.

Table 6: *Simulator parameters.*

Parameter	Value	Notes
Earth cross-cont. delay	30–120 ms	per link pair
LEO–ground delay	20–45 ms	per ground station
Moon one-way delay	1.28 s	fixed
Mars one-way delay	186–1342 s	swept
Mars-local delay	5 ms	fixed
Processing delay	1 ms	per acceptor
Jitter	$\pm$10%	of link delay, uniform
Per-phase timeout	500 s / 1 s	global / tier-local; global budget raised to 1.25$\times$ the worst-case request–response path when that exceeds the base (above $\approx$200 s Mars delay)
Max Paxos rounds	1	per attempt
Blackout start	600 s base	pre-window scaled to fit one full global round plus margin (930 s at 186 s Mars delay)
Simulation end	4000 s base	horizon extends when pre/post windows scale
Reconciliation interval	120 s	between global attempts
Loss model	none	deterministic link removal for blackout

Table 7: *Claim-to-artifact traceability.*

Claim	Evidence	Artifact(s)
Flat construction: 0% during blackout at every hard-blackout sweep point.	Flat-mode sweep under the validated harness (`--global-quorum flat`).	`results/step9_flat/step9_sweep.csv`; `experiments/step9_sweep.py`.
Majority baseline: 100% during blackout, 361 ms two-phase decision latency at every hard-blackout sweep point.	Majority-mode sweep under the validated harness (`--global-quorum majority`).	`results/step9_maj/step9_sweep.csv`; `experiments/step9_sweep.py`.
Crumbling wall: 100% during blackout at every hard-blackout sweep point, 183 ms two-phase decision latency.	All rows show 100% during-blackout.	`results/step9/step9_sweep.csv`; `experiments/step9_sweep.py`.
Per-tier liveness: 3 of 4 tiers retain global consensus during blackout.	Per-tier sweep, both topologies.	`results/tier_liveness/tier_sweep_full_ci.csv`; `experiments/tier_liveness_sweep.py`.
Sparse topology: LEO drops to 0%.	Sparse rows in per-tier sweep.	Same as above.
Weakest-link migration and coordinated relaxation.	Crash-tolerance sweep rows.	`results/step10/step10_sweep_ci.csv`; `experiments/step10_sweep.py`.
TLA+ verification of quorum intersection.	Exhaustive enumeration, all four tiers $\times$ $\{$strict, k=4, k=3$\}$ (27,921 states, complete).	`tla/QuorumIntersection.tla`; `tla/ExhaustiveIntersection.tla`.
Paxos agreement model-checked.	67M states (reduced topology, all-tier Phase 1 family, complete).	`tla/PaxosSmall.tla`.
Per-tier capability map: which (Phase 1, Phase 2) capability states each tier holds under each connectivity regime.	Exhaustive per-scenario classification, including the degraded partial-capability states.	`results/capability/capability_map.csv`; `experiments/capability_map.py`.
Containment characterization (Prop. 2) and the $k \leq \lceil \lvert E \rvert /2 \rceil$ boundary (Prop. 3).	Exhaustive enumeration, 5 k $\times$ 4 tiers; containment evaluated as set containment over minimal Q_2, not as the reachability restatement; 0 disagreements. Predicate monotonicity verified, not assumed.	`results/capability/dual_containment.csv`; `experiments/capability_dual_sweep.py`.
Uniform threshold corollary: $(1,0)$ reachable iff $q_1 < q_2$; $(0,1)$ iff $q_2 < q_1$.	80 threshold configurations ($n = 3..7$, all valid q_1/q_2), 0 violations.	`results/capability/dual_uniform.csv`; `experiments/capability_dual_sweep.py`.
Valence of the two capability states: $(1,0)$ indistinguishable from a healthy contender; $(0,1)$ blocks the healthy proposer at retry budget 8.	Pre-registered experiment, 5 arms $\times$ 4 retry budgets $\times$ 50 seeds; predictions, falsification criterion, and declared consequence of a null committed before the experiment code existed; two registered predictions falsified and kept. Output byte-identical across two runs.	`results/flip/flip_map.csv`; `results/flip/flip_sweep.csv`; `flip.py`; `experiments/flip_sweep.py`; `experiments/flip_verdict.py`.
Price of coincidence: 0 at odd n, 1 at even n; cost-minimal configurations cannot be phase-symmetric at even n.	Re-analysis of the threshold enumeration; post-hoc, not pre-registered, and labelled as such.	`results/capability/dual_uniform.csv`; `experiments/capability_dual_sweep.py`.
Tier-gradient: no k closes both capability states; T_2 and T_3 retain $(0,1)$ at every k.	Exhaustive enumeration over all 2^{10} connectivity states, every tier and k, reported under both colocated-acceptor readings. Output byte-identical across two runs.	`results/capability/dual_gradient_map.csv`; `experiments/capability_dual_sweep.py`.